

Executive Summary

This project focuses on **analyzing website traffic** and music interaction data to understand user behavior, engagement patterns, and conversion opportunities for Alfido Tech. The dataset was obtained from Kaggle and contained user interaction events such as clicks, previews, and downloads along with artist, track, and geographic information.

The project involved data cleaning, preprocessing, exploratory data analysis, and feature engineering to generate synthetic session-level analytics such as user IDs, session IDs, session duration, landing pages, exit pages, and referral sources. Using Python libraries including Pandas, Matplotlib, Seaborn, Plotly, and NetworkX, multiple visualizations and traffic metrics were generated.

Key findings revealed that click events dominated user activity, a small number of artists and tracks generated the majority of engagement, and traffic was concentrated in specific countries and cities. The analysis also identified user drop-offs within the conversion funnel and highlighted opportunities to improve user retention and conversion rates.

Based on the analysis, several recommendations were proposed for Alfido Tech, including optimizing landing pages, improving the conversion funnel, promoting high-performing artists, targeting high-engagement regions, and enhancing user navigation experience. The project demonstrates how data analytics and traffic analysis can support better business decisions and improve digital platform performance.

Code Links

[https://github.com/satyamkumari/Alfido-Tech-Internship/blob/main/Website traffic analysis/website traffic analysis.ipynb](https://github.com/satyamkumari/Alfido-Tech-Internship/blob/main/Website%20traffic%20analysis/website%20traffic%20analysis.ipynb)

Screenshot & Visualization

6 — BOUNCE RATE PIE CHART

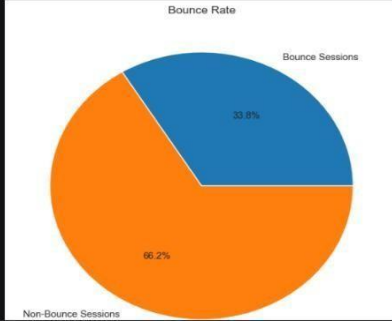
```
plt.figure(figsize=(7,7))

labels = [
    'Bounce Sessions',
    'Non-Bounce Sessions'
]

sizes = [
    bounces,
    total_sessions - bounces
]

plt.pie(
    sizes,
    labels=labels,
    autopct='%1.1f%%'
)

plt.title("Bounce Rate")
plt.show()
```

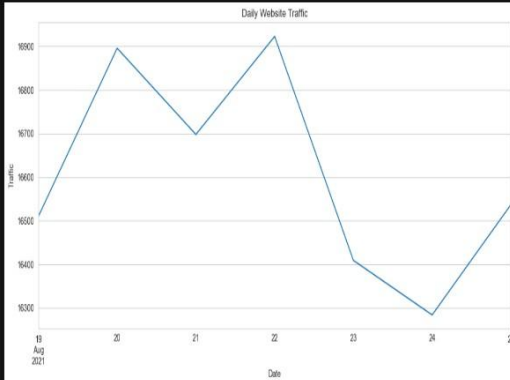


5 — DAILY TRAFFIC TREND

```
plt.figure(figsize=(14,4))

daily_traffic.plot()

plt.title("Daily Website Traffic")
plt.xlabel("Date")
plt.ylabel("Traffic")
plt.show()
```

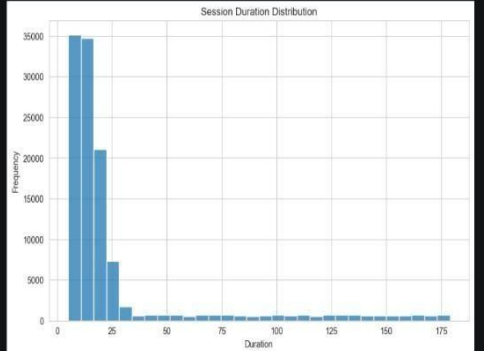


12 — SESSION DURATION DISTRIBUTION

```
plt.figure(figsize=(10,6))

sns.histplot(
    df['duration'],
    bins=30
)

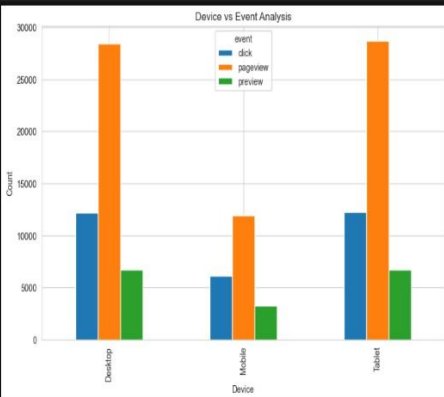
plt.title("Session Duration Distribution")
plt.xlabel("Duration")
plt.ylabel("Frequency")
plt.show()
```



DEVICE ANALYSIS

```
device_analysis.plot(
    kind='bar',
    figsize=(10,6)
)

plt.title("Device vs Event Analysis")
plt.xlabel("Device")
plt.ylabel("Count")
plt.show()
```



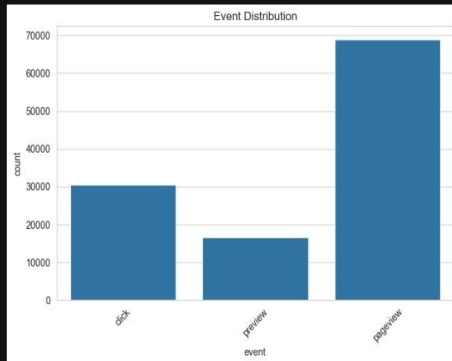
VISUALIZATIONS

1 — EVENT DISTRIBUTION

```
plt.figure(figsize=(8,5))

sns.countplot(
    x='event',
    data=df
)

plt.title("Event Distribution")
plt.xticks(rotation=45)
plt.show()
```



11 — USER FLOW GRAPH

```
u = ux.from_pandas_adjlist(
    ux_graph,
    source='page',
    target='next_page',
    edge_attr='event',
    create_using=ux.DiGraph()
)

plt.figure(figsize=(11,8))

pos = nx.spring_layout(u)

nx.draw(
    u,
    pos,
    with_labels=True,
    node_size=400,
    font_size=10
)

plt.title("User Navigation Flow")
plt.show()
```

